King Abdulaziz University

College of Engineering

Electrical and Computer Engineering

EE 424 (Antenna)

Project Part 1

(Microstrip Transmission Line)

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Objective: Design and simulate a 50Ω microstrip transmission line with 1.2GHz and parameters values given by instructor substrate material FR4 (εr = 4.3), and thickness (0.7) with electrical length (pi/4).

Starting with calculations we need to calculate effective dielectric constant so we can get the right length:

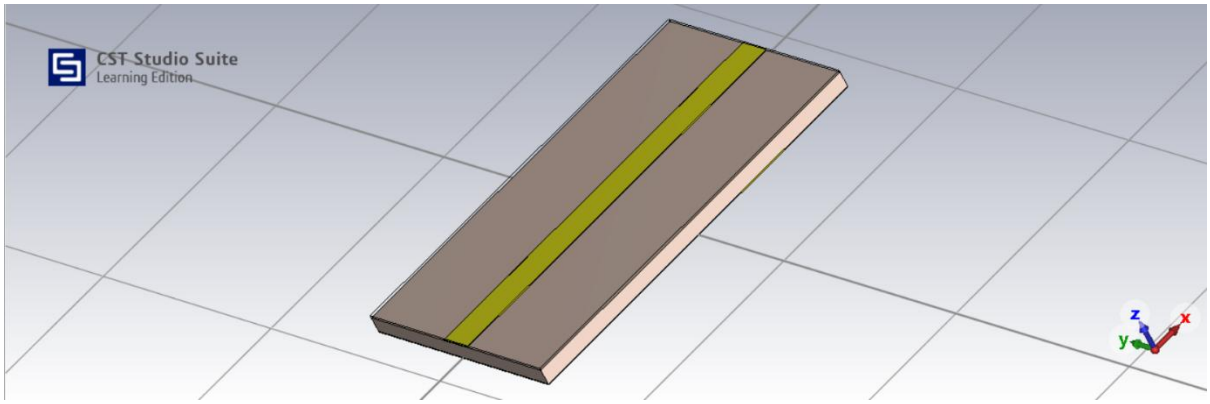
Hight = 0.7

Width = 1.36mm

Length = 17.293mm

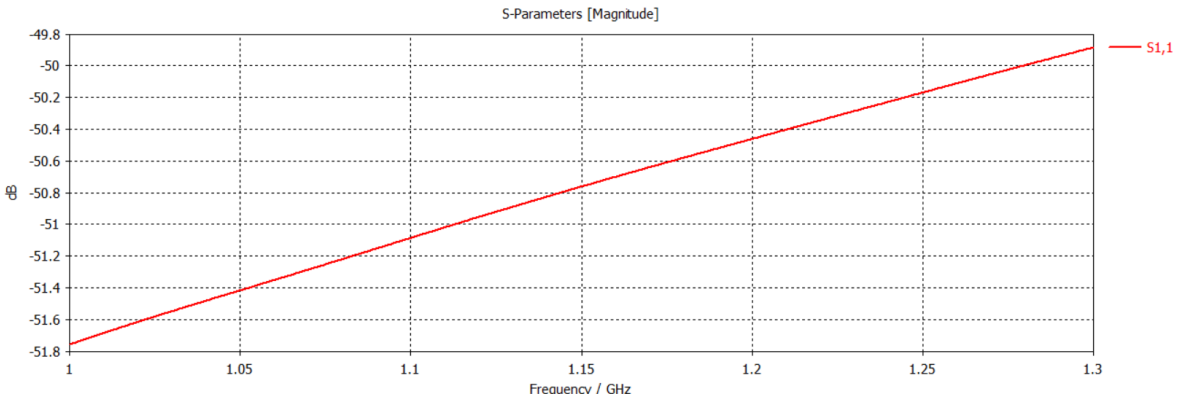
effective dielectric constant = 3.265

All the calculations performed by hand on last page**

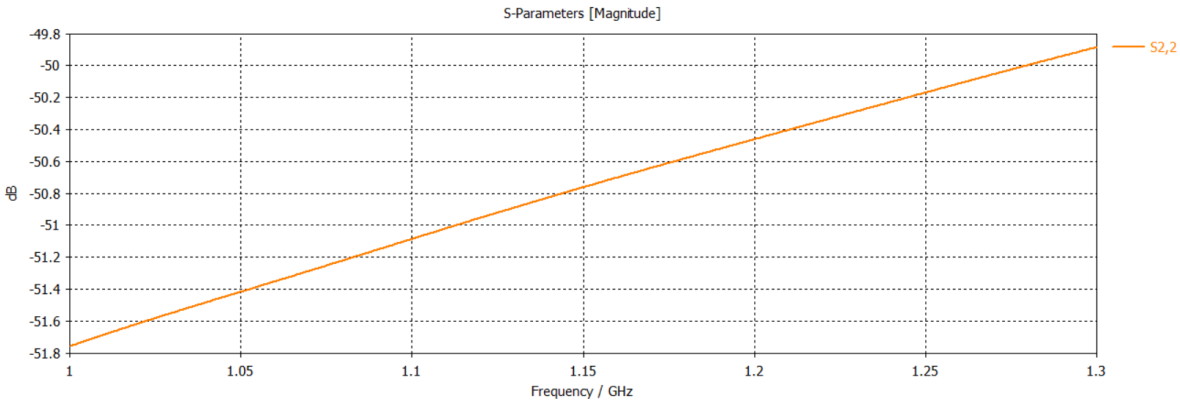


Microstrip Transmission Line

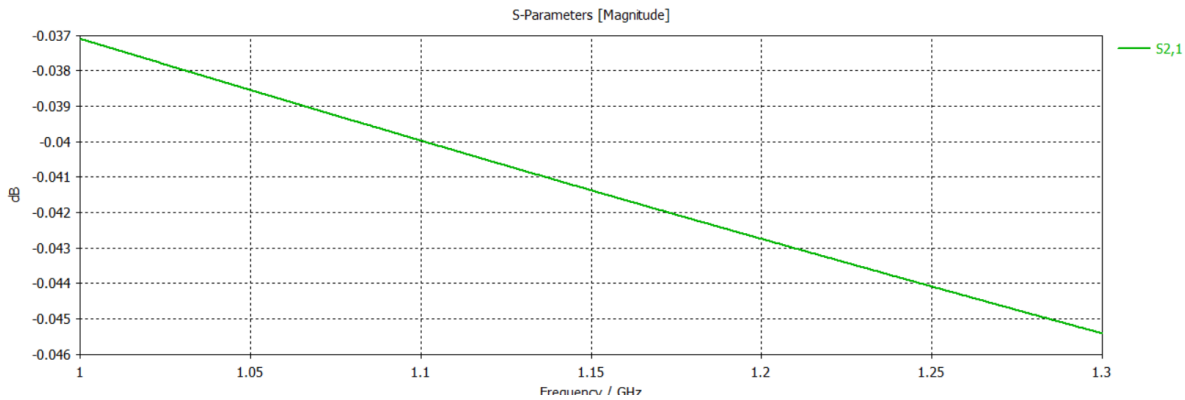
After designing and simulating microstrip line we got S11 and S22 very low around -54 dB and that shows perfect transmitting power with no loses. S12 and S21 close to 0dB indicate strong power transmitting. All results show an ideal performance of matched microstrip TL.



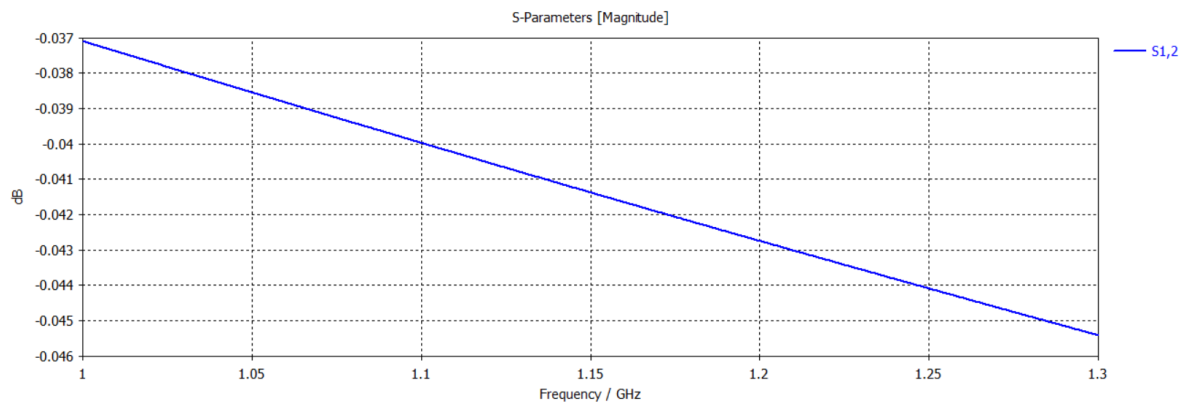
S11



S22


S-Parameters [Magnitude]
-0.037
-0.038
-0.039
-0.04
-0.041
-0.042
-0.043
-0.044
-0.045
-0.046
dB
1
1.05
1.1
1.15
1.2
1.25
1.3
Frequency / GHz
S2,1

S21


S-Parameters [Magnitude]
-0.037
-0.038
-0.039
-0.04
-0.041
-0.042
-0.043
-0.044
-0.045
-0.046
dB
1
1.05
1.1
1.15
1.2
1.25
1.3
Frequency / GHz
S1,2

S12

References:

1- Lecture 16

2- Calculations done by hand:

L16 P1

$h = 0.7\,mm \qquad \varepsilon_r = 4.3 \qquad \phi = \frac{\pi}{4} \qquad f = 1.2\,GHz \qquad Z_0 = 50$

$$A = \frac{50}{60}\sqrt{\frac{\varepsilon_r+1}{2}} + \frac{\varepsilon_r-1}{\varepsilon_r+1}\left(0.23 + \frac{0.11}{\varepsilon_r}\right) = 1.356 + 0.159 = 1.515$$

$$B = \frac{377\pi}{2 \times 50\sqrt{\varepsilon_r}} = 5.105$$

if

$$2 > \frac{W}{h} = \frac{8e^{A}}{e^{2A}-2} = \frac{8e^{1.515}}{e^{2.1515}-2} = 1.946 < 2 \checkmark$$

$$W = 1.946\,h = 1.946 \times 0.7\,mm = 1.36\,mm$$

$$\varepsilon_e = \frac{\varepsilon_r+1}{2} + \frac{\varepsilon_r-1}{2}\left[\frac{1}{\sqrt{1+12\frac{h}{W}}}\right] = 3.265 \qquad \lambda_g = \frac{c}{f\sqrt{\varepsilon_e}} = \frac{3\times10^8}{1.2\times10^9 \times \sqrt{3.265}} = 138.35$$

$$\beta = \frac{2\pi}{\lambda_g} = \frac{2\pi}{138.35\,mm} = 45.415$$

$$l = \frac{\phi}{\beta} = 17.293\,mm$$